Лекция 13. Машинный перевод. Механизм внимания. Трансформеры

Фролов Дмитрий

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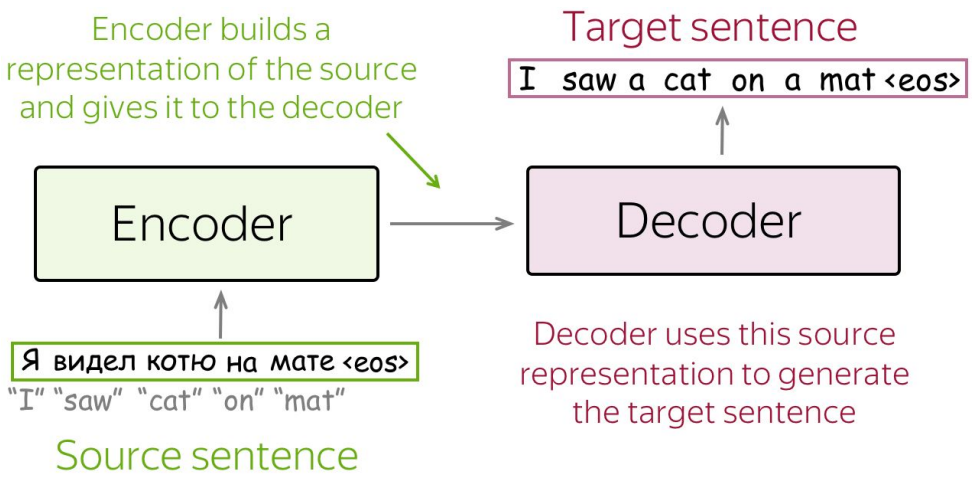
Машинный перевод как языковая модель

Language Models: $P(y_1, y_2, \dots, y_n) = \prod_{t=1}^{n} p(y_t | y_{<t})$

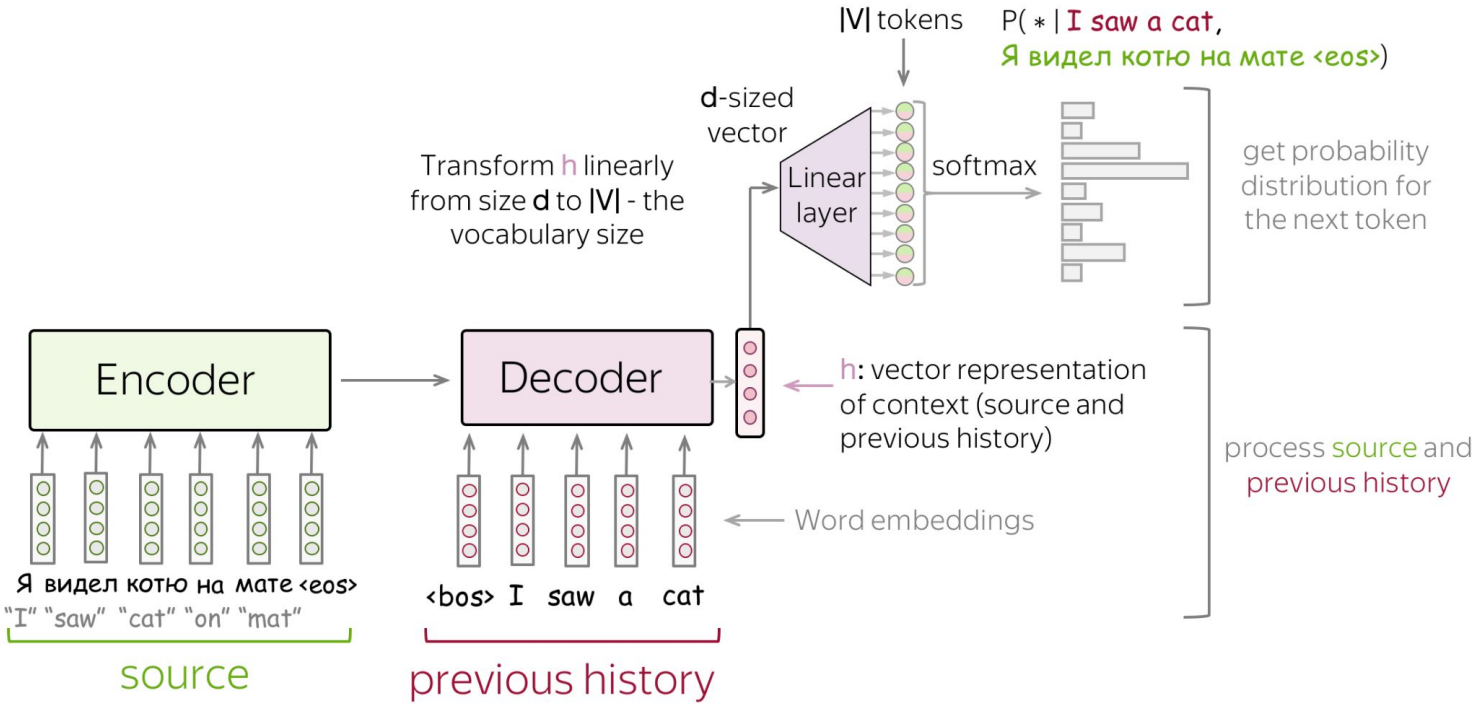
Conditional Language Models: $P(y_1, y_2, \dots, y_n, | x) = \prod_{t=1}^{n} p(y_t | y_{<t}, x)$

condition on source x



Images Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

Архитектура *seq2seq*



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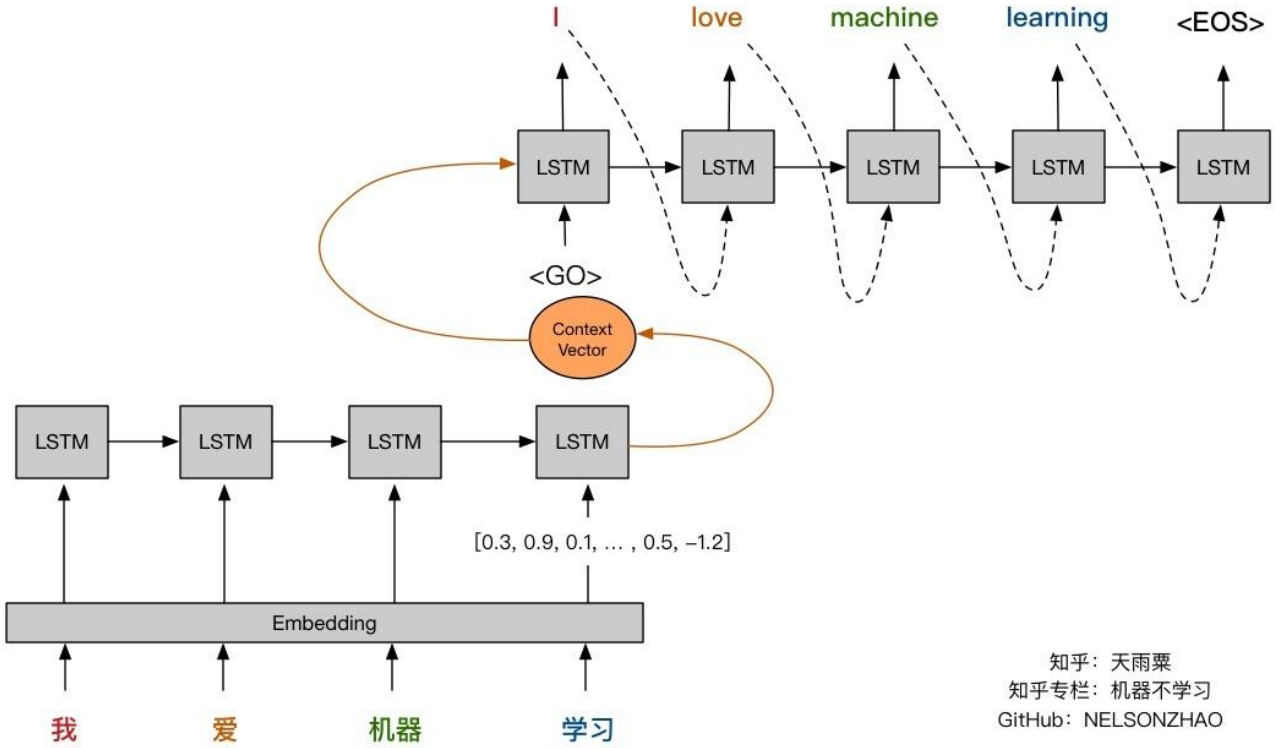
seq2seq на основе RNN



Sutskever I et al. Sequence to sequence learning with neural networks. Advances in neural information processing systems. 2014;27.

Механизм внимания

Attention output

$$c^{(t)} = a_1^{(t)} s_1 + a_2^{(t)} s_2 + \cdots + a_m^{(t)} s_m = \sum_{k=1}^{m} a_k^{(t)} s_k$$

"source context for decoder step t"

(weighted sum)

Attention weights

$$a_k^{(t)} = \frac{\exp(\text{score}(h_t, s_k))}{\sum_{i=1}^{m} \exp(\text{score}(h_t, s_i))}, \mathrm{k} = 1..\mathrm{m}$$

"attention weight for source token k at decoder step t"

(softmax)

Attention scores

$$\text{score}(h_t, s_k), \mathrm{k} = 1..\mathrm{m}$$

"How relevant is source token k for target step t?"

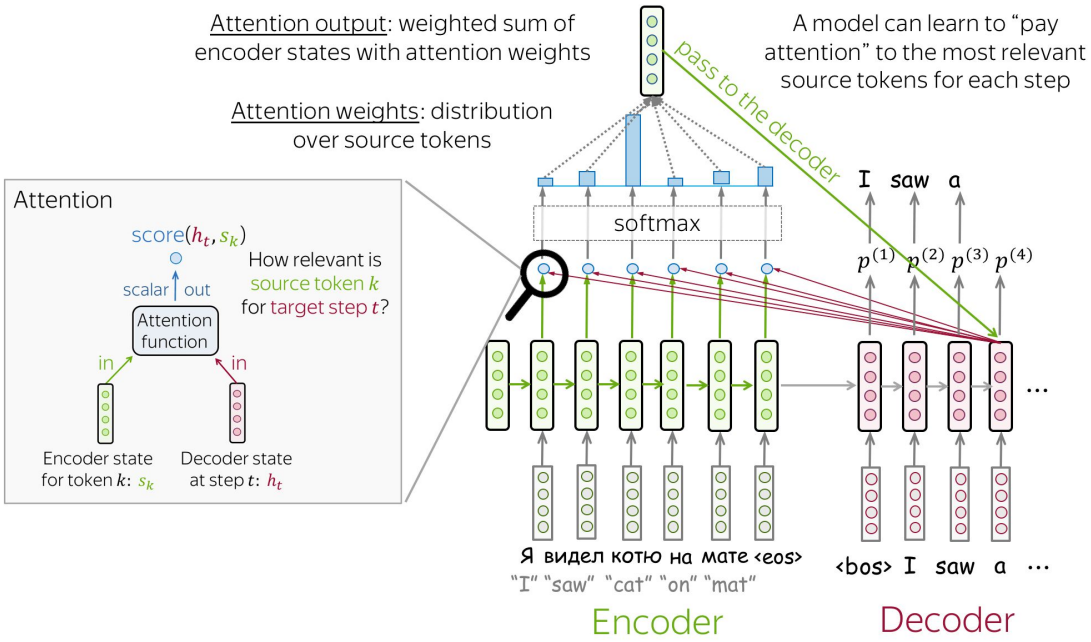
Attention input

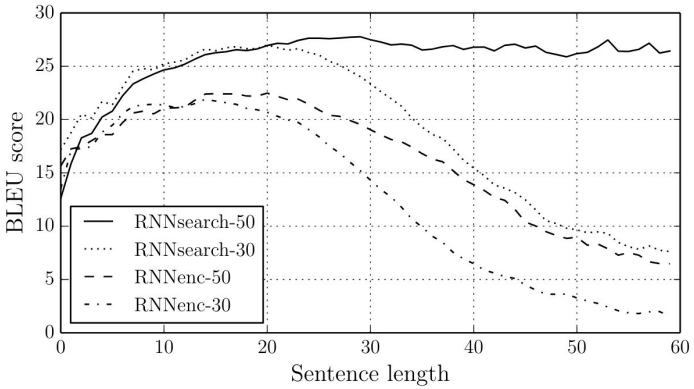
$s_1, s_2, \ldots, s_m$ — all encoder states

h_t — one decoder state

Image Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

seq2seq с вниманием

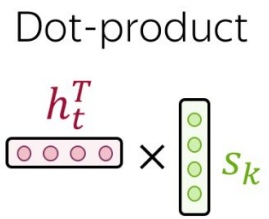




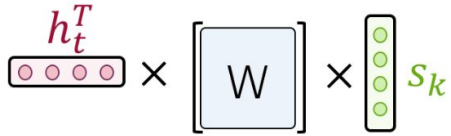
Bahdanau D et al. Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473. 2014 Sep 1.

Image Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

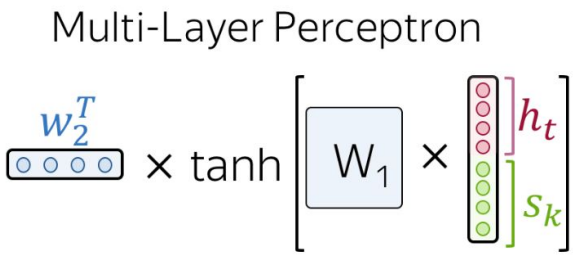
Способы расчета внимания



$$\text{score}(h_t, s_k) = h_t^T \; s_k$$



$$\text{score}(h_t, s_k) = h_t^T \; W s_k$$



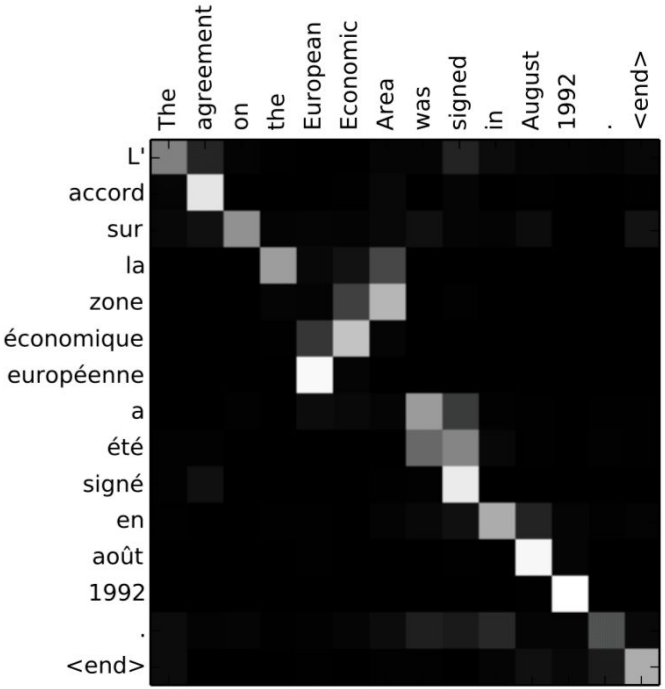
$$\text{score}(h_t, s_k) = w_2^T \cdot \tanh(W_1[h_t, s_k])$$

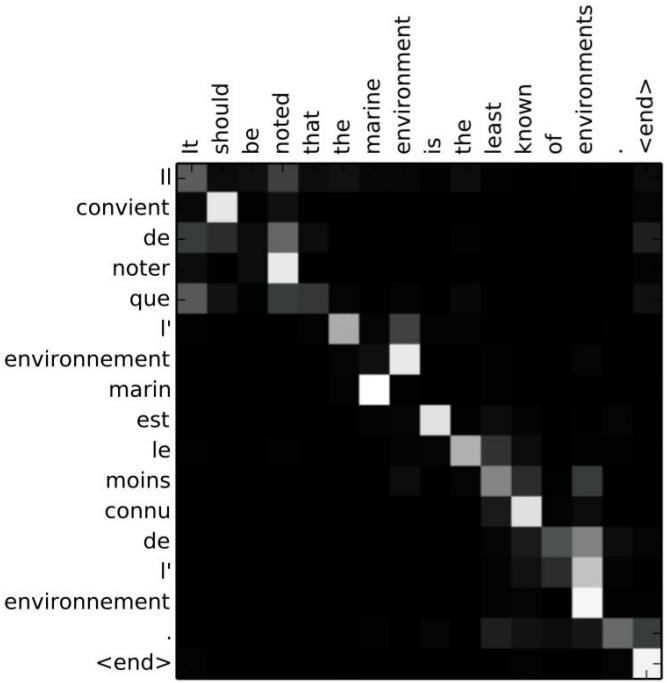
Luong MT et al. Effective approaches to attention-based neural machine translation. InProceedings of the 2015 conference on empirical methods in natural language processing 2015 Sep (pp. 1412-1421).

Bahdanau D et al. Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473. 2014 Sep 1.

Image Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

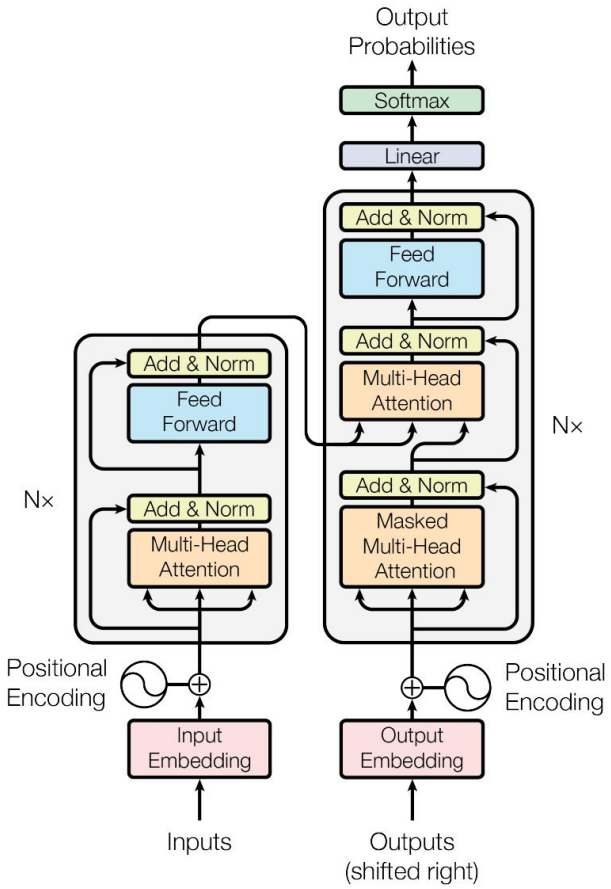
Интерпретация внимания





Bahdanau D et al. Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473. 2014 Sep 1.

Архитектура трансформера



Vaswani A et al. Attention is all you need. Advances in neural information processing systems. 2017;30.

QKV-Attention

Each vector receives three representations ("roles")

$[W_Q] \times \square = \square$ Query: vector **from** which the attention is looking

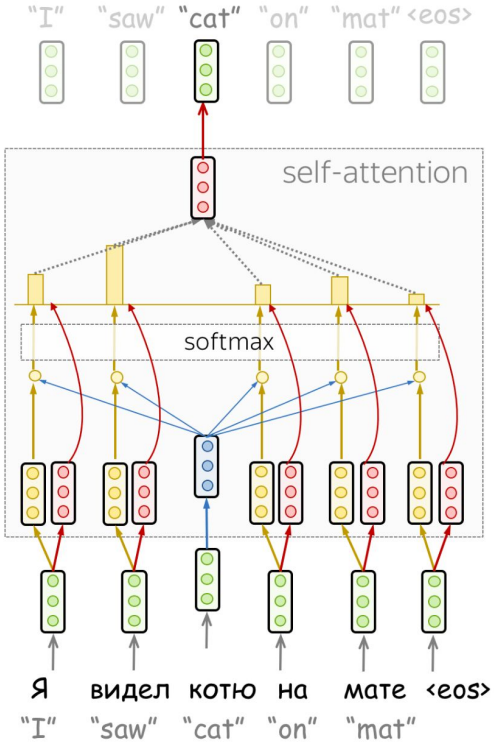
"Hey there, do you have this information?"

$[W_K] \times \square = \square$ Key: vector **at** which the query looks to compute weights

"Hi, I have this information – give me a large weight!"

$[W_V] \times \square = \square$ Value: their weighted sum is attention output

"Here's the information I have!"



$$Attention(q, k, v) = \overbrace{softmax\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} v$$

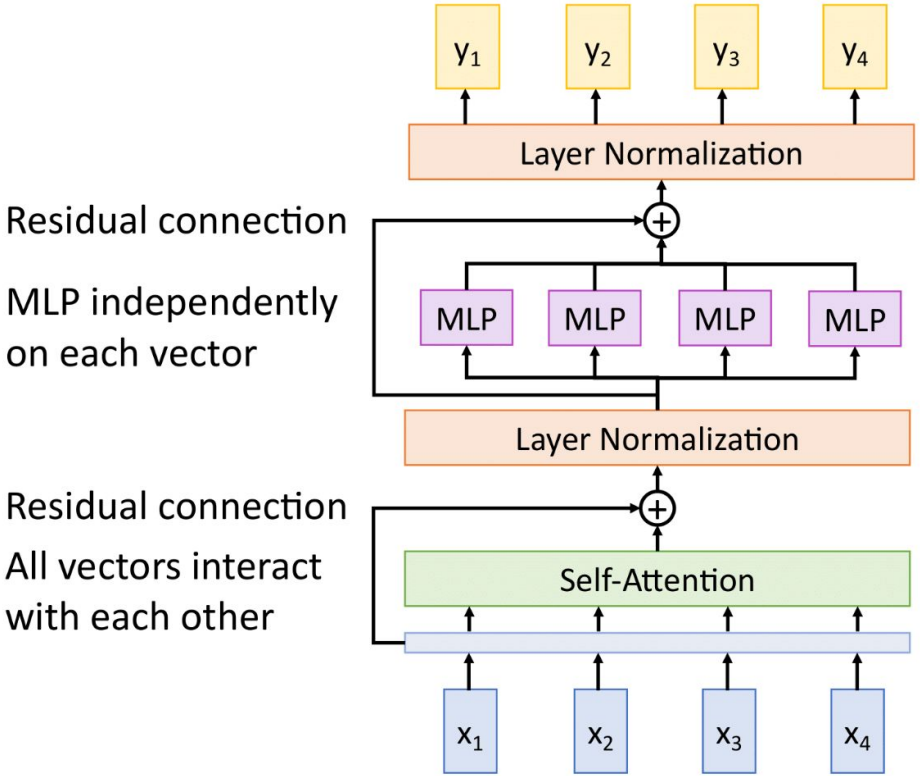
from to

vector dimensionality of K, V

Vaswani A et al. Attention is all you need. Advances in neural information processing systems. 2017;30.

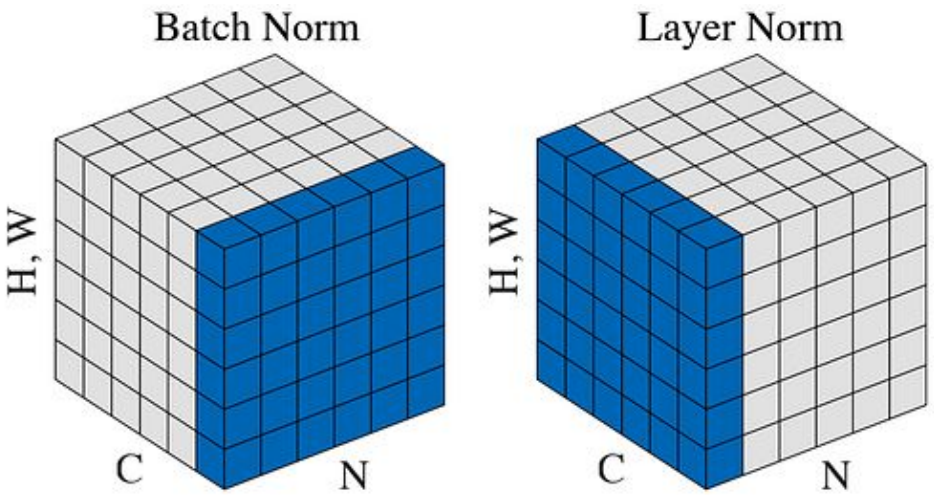
Images Source: https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html

Блок кодировщика трансформера

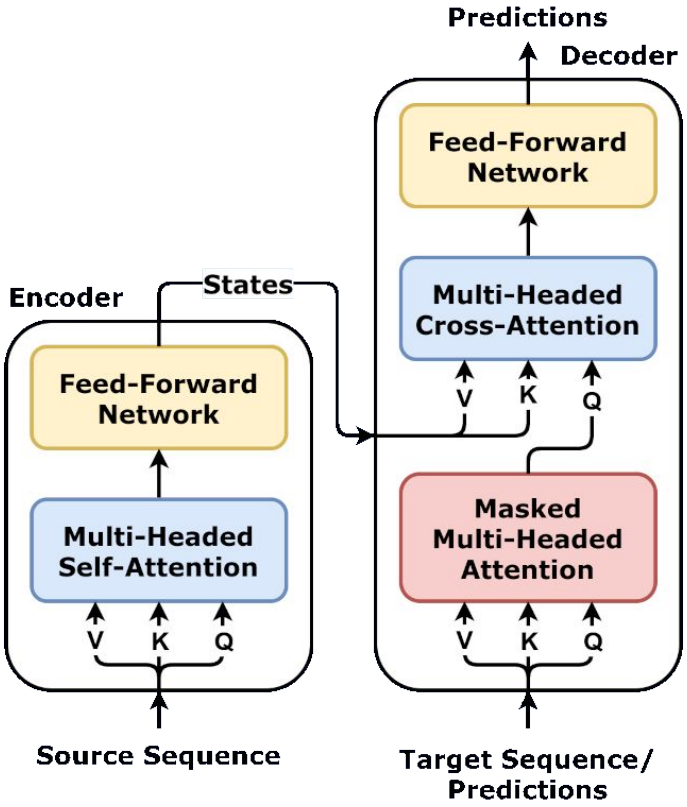


Vaswani A et al. Attention is all you need. Advances in neural information processing systems. 2017;30.

Batch vs Layer Norm

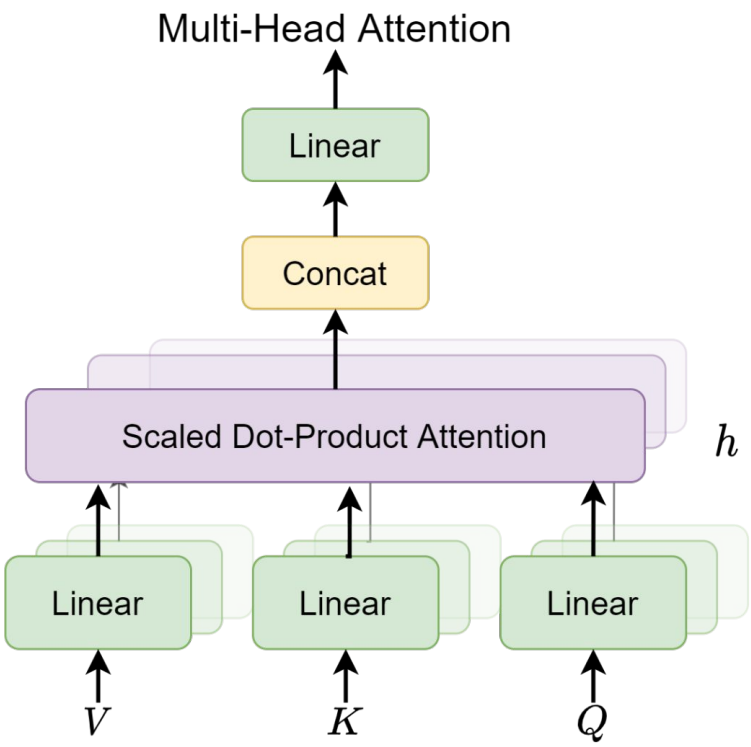


Кодировщик vs Декодировщик



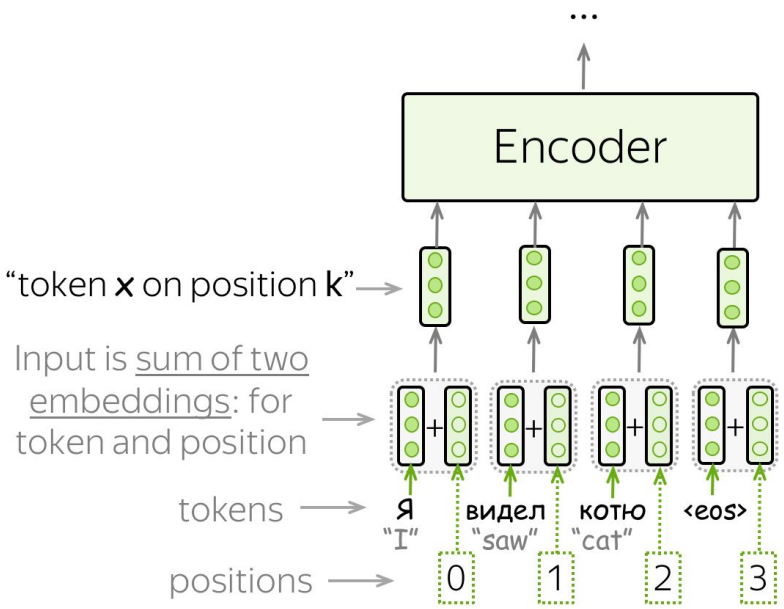
Vaswani A et al. Attention is all you need. Advances in neural information processing systems. 2017;30.

Multi-Head



Vaswani A et al. Attention is all you need. Advances in neural information processing systems. 2017;30.

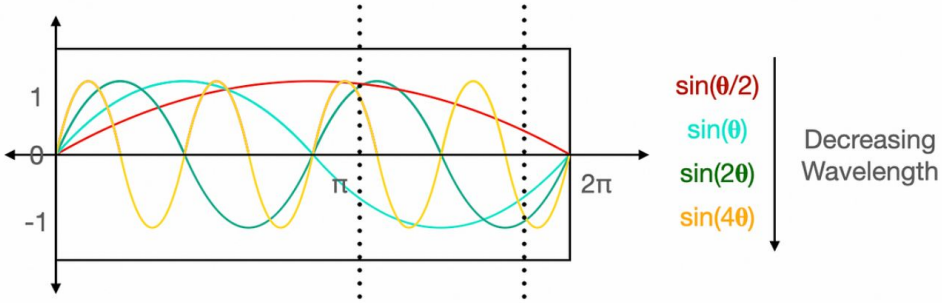
Позиционное кодирование



Формула:

- $PE_{pos,2i} = \sin(pos/10000^{2i/d})$
- $PE_{pos,2i+1} = \cos(pos/10000^{2i/d})$

где pos — позиция, i — индекс размерности, d — размер эмбеддинга.



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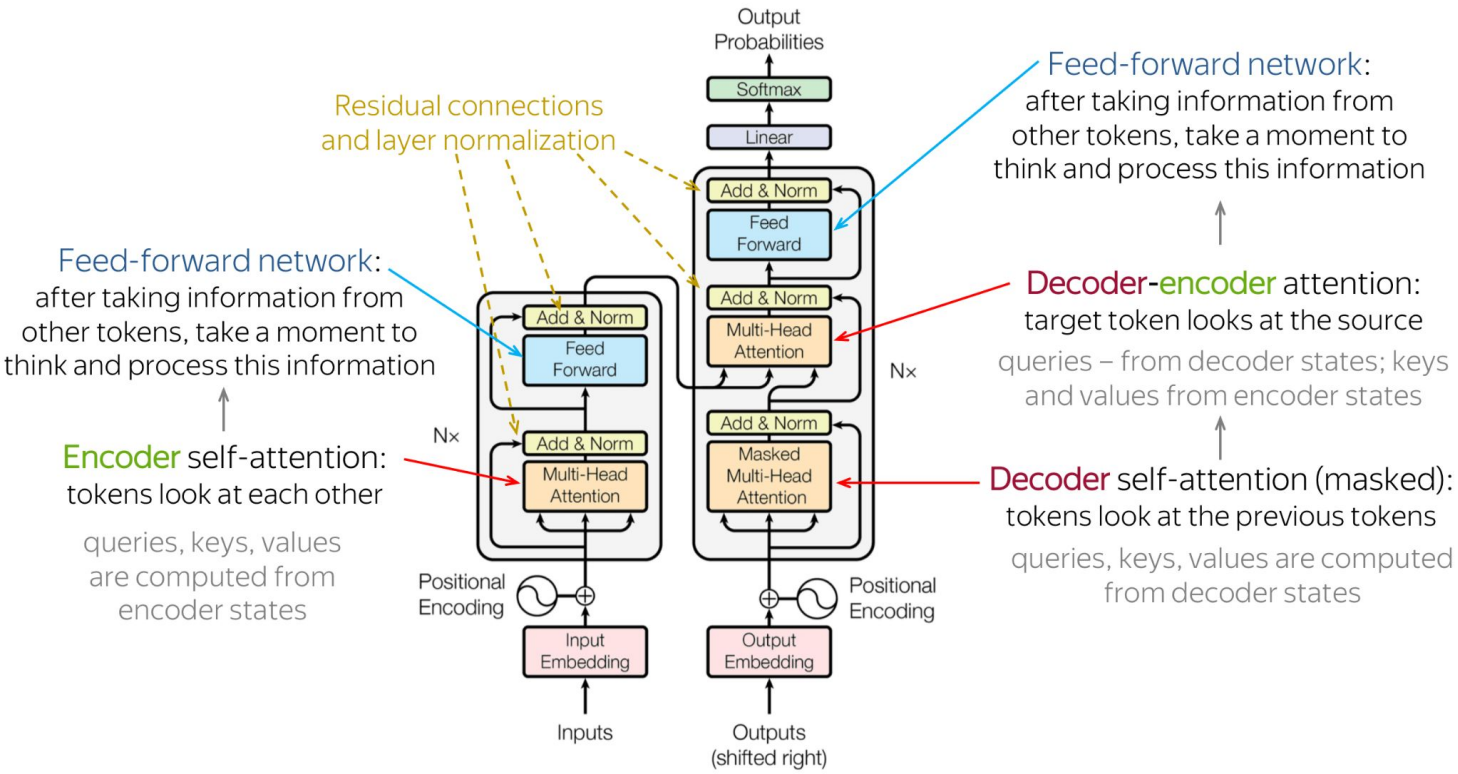
Подытожим



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